Google Developer Group
Universitas Sriwijaya

Learning 7

Dynamic Routing & Single Page Navigation

Navigation & Routing in React Applications

Why Routing Matters

- Single Page Apps still need navigation
- URL represents application state
- Different URLs → different UI
- Improves usability and shareable links

```
/            → Home
/products    → Product List
/products/1  → Product Detail
```

React Router Mental Model

- React Router maps URL → UI
- Routing is defined with components
- Routes describe application structure
- UI updates automatically when URL changes

```
<Route path="/about" element={<About />} />
```

Basic Router Setup

- BrowserRouter enables routing
- Usually wraps the root of the app
- Provides routing context to components

```
import { BrowserRouter } from "react-router-dom"

<BrowserRouter>
  <App />
</BrowserRouter>
```

Defining Routes

- Routes contains route definitions
- Route maps *path* → *component*
- Determines which UI appears for a URL

```
<Routes>
  <Route path="/" element={<Home />} />
  <Route path="/about" element={<About />} />
</Routes>
```

Nested Routes

- Routes can be nested
- Useful for complex UI structures
- Child routes render inside parent routes

```
/dashboard
/dashboard/settings
```

```
<Route path="dashboard" element={<Dashboard />}>
  <Route path="settings" element={<Settings />} />
</Route>
```

Layout Routes

- Share layout between multiple pages
- Layout contains common UI
- Example: navbar, sidebar, footer

```
<Route element={<Layout />}>
  <Route path="/" element={<Home />} />
  <Route path="/about" element={<About />} />
</Route>
```

Index Routes

- Default route inside a parent route
- Rendered when no child path matches
- Used for default page in a section

```
<Route path="dashboard" element={<Dashboard />}>
  <Route index element={<Overview />} />
</Route>
```

Dynamic Routes

```
<Route path="projects">
  <Route path=":id" element={<Project />} />
</Route>
```

```
/projects/42
```

Route Params

```
const { id } = useParams()
```

Navigation with Link

- Navigate between routes
- Prevents full page reload
- Uses client-side routing
- Works like an anchor tag

```
import { Link } from "react-router-dom"

<Link to="/dashboard">Dashboard</Link>
<Link to="/about">About</Link>
```

NavLink

- Special version of Link
- Detects the active route
- Useful for navigation menus
- Can apply active styling

```
import { NavLink } from "react-router-dom"

<NavLink to="/dashboard">
  Dashboard
</NavLink>

<NavLink
  to="/dashboard"
  className={({ isActive }) =>
    isActive ? "text-blue-600 font-bold" : ""
  }
>
  Home
</NavLink>
```

Challenge! (FE Member Only*)

1. Eksplorasi dan praktikkan jenis-jenis kasus Routing

Selamat Mencoba ~

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Thank You !